The Campus Premiere Ticket Counter

Week 8 Coding Assignment - Question 3

Design: Customer 1--0..* Booking; Show 1--0..* Booking. Booking has 1..6 seats.
Show reserves unique seat IDs. A SeatType strategy provides its category-specific price.
Java solution:

```
import java.util.*;
interface SeatType { double price(); }
class Regular implements SeatType { public double price(){return 150;} }
class Premium implements SeatType { public double price(){return 250;} }
class Recliner implements SeatType { public double price(){return 400;} }
class Seat { final String id; final SeatType type; Seat(String i,SeatType t){id=i;type=t;} }
class Customer { final String name; Customer(String n){name=n;} }
class Show { final Set<String> booked=new HashSet<>(); boolean started=false; }
class Booking {
final Customer customer; final Show show; final List<Seat> seats; boolean cancelled=false;
Booking(Customer c,Show s,List<Seat> ss){customer=c;show=s;seats=new ArrayList<>(ss);}
double total(){return seats.stream().mapToDouble(s->s.type.price()).sum();}
boolean cancel(){if(cancelled||show.started)return false;for(Seat
s:seats)show.booked.remove(s.id);cancelled=true;return true;}
}
class TicketDesk {
Booking book(Customer c,Show show,List<Seat> seats){
if(seats.isEmpty()||seats.size()>6)throw new IllegalArgumentException("Choose 1 to 6 seats");
Set<String> wanted=new HashSet<>();for(Seat s:seats){if(!wanted.add(s.id))throw new
IllegalArgumentException("Duplicate seat "+s.id);if(show.booked.contains(s.id))throw new
IllegalStateException("Seat "+s.id+" is already booked for this show.");}
show.booked.addAll(wanted);return new Booking(c,show,seats);
}
}
public class Main { public static void main(String[] args){
TicketDesk d=new TicketDesk();Show show=new Show();Customer a=new Customer("Asha"),r=new
Customer("Ravi"),n=new Customer("Neha");
Booking b=d.book(a,show,Arrays.asList(new Seat("A1",new Regular()),new Seat("A2",new
Regular()),new Seat("F5",new Premium())));
System.out.printf("Booking confirmed for Asha: A1, A2, F5. Total: Rs %.2f.%n",b.total());
try{d.book(r,show,Arrays.asList(new Seat("A2",new Regular())));}catch(IllegalStateException
e){System.out.println(e.getMessage());}
Booking rb=d.book(r,show,Arrays.asList(new Seat("R1",new
Recliner())));System.out.printf("Booking confirmed for Ravi: R1. Total: Rs
%.2f.%n",rb.total());
if(b.cancel())System.out.println("Asha’s booking cancelled. Seats A1, A2, F5 released.");
Booking nb=d.book(n,show,Arrays.asList(new Seat("A2",new
Regular())));System.out.printf("Booking confirmed for Neha: A2. Total: Rs
%.2f.%n",nb.total());
} }
```

Expected totals: Asha Rs 550.00; Ravi Rs 400.00; Neha Rs 150.00.
Cancellation is rejected once show.started is true and releases every seat for a valid cancellation.